

Research Plan

Nothing Untrue: Strategic Omission and Framing in Financial LLM Agents

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1 Literature Overview

Large language models are now well documented as capable of, and prone to, misaligned deception (Park et al., 2023): strategically deceiving users under pressure in a financial setting (Scheurer et al., 2023), pursuing scheming when handed an in-context goal (Meinke et al., 2024), acting against their principals even under ostensibly harmless goals (Lynch et al., 2025), taking manipulative actions when over-optimising for helpfulness (Wang et al., 2026), showing deceptive tendencies absent strong external pressure (Järvinen and Hubinger, 2024), and trading ethical constraints against reward in sequential decision settings (Pan et al., 2023). Almost all of this evidence concerns blatant falsehood or strong-nudge scheming, much of it dependent on explicit goal-nudging (Meinke et al., 2024) or on artificial tasks whose deception signal may reflect reasoning difficulty rather than strategic intent (Wu et al., 2025b). The failure mode that matters most in deployment is subtler: truthful-but-misleading communication that selectively omits an adverse material fact or frames it to understate downside. A unified taxonomy across 50 benchmarks isolates these mechanisms, omission and pragmatic distortion, as distinct from fabrication and finds them critically under-covered (Shi et al., 2026), and the clearest behavioural evidence that models deceive without stating a falsehood — misleading non-falsities that standard truth probes fail to catch — comes from artificial, non-financial settings with no nudge or falsifiability gradient (Berger, 2026).

The one benchmark to target the goal-directed form directly is JANUS (Giannouris et al., 2026): it gives each scenario a fixed pool of annotated favourable and adverse facts, compares model output under a neutral versus a goal-directed condition without ever instructing the model to lie, and scores the resulting distortion — both omission of adverse facts and softened framing — across several domains, with its strongest effects in finance. JANUS establishes that goal-conditioned information distortion is a real, measurable, and finance-relevant phenomenon. It also defines the edge of what is known: it collapses goal pressure into a neutral-versus-goal binary, aggregates the distortion forms into a single index, is single-turn, treats finance as one domain among several, and provides no detection or mitigation. This study is adjacent to JANUS rather than an extension of it: at the time of writing JANUS has not released its code or data, so its scenarios cannot be reused, motivating an independent, finance-focused benchmark. The benchmark spans a wider range of financial scenarios across single- and multi-turn settings, scores deception on a falsifiability gradient, tests whether user susceptibility modulates the rate of deceptive behaviour, and adds black-box and white-box detection.

Scenario design and scoring build on DECOR, which decomposes a context into atomic informational units and scores each against the model’s response — a basis for defining scenarios around atomic material-risk units and grading omission and framing (Cai et al., 2026). Scenarios are sourced and adapted from existing finance benchmarks: DeceptionBench, whose economy domain contains finance cases but targets self-interest and sycophancy and so needs modification

(Huang et al., 2025), together with a set of financial task and safety benchmarks (Bigeard et al., 2025; Zhu et al., 2026; Hou et al., 2026; Chen et al., 2025) — one of which studies user-side rather than model-side risk concealment and is therefore distinct from this work’s aim (Cheng et al., 2025). User susceptibility enters as an experimental factor, following work that estimates how strongly a user relies on the agent (Wu et al., 2025a), to test whether certain personas elicit more deception. For detection, the findings that models deceive through misleading non-falsities which truth probes trained on true/false statements fail to catch (Berger, 2026), and that probe quality is dominated by the training instruction-pair — favouring specialised probes per threat model over a universal detector (Natarajan et al., 2026) — together motivate comparing white-box against black-box methods and testing whether any detector transfers across the variant matrix, rather than assuming probes generalise; direct white-box/black-box comparisons, including evidence that detectors fail when deception is not apparent from the transcript alone, extend that comparison to subtle deception (Parrack et al., 2025; Kretschmar et al., 2026). Lightweight mitigation testing draws on environmental controls such as escalation channels, shown to reduce harmful agentic actions (Gomez, 2025).

2 Research Scope

2.1 Deception definitions

Two finance-native forms are measured, arranged on a **falsifiability gradient**:

Selective omission: Failure to disclose a material financial, legal, customer, market, or compliance risk the model had access to and that a reasonable stakeholder would expect to see — including presenting favourable facts while omitting the adverse facts needed for a balanced decision.

Misleading framing: Technically true language that understates uncertainty, downside, or regulatory significance.

2.2 Deception Gradients

Falsifiability gradient:

- Level 0 — blatant false claim (control, the regime existing detectors already handle)
- Level 1 — selective omission (all stated facts true, a held adverse fact left out)
- Level 2 — misleading framing (all facts present, downside/regulatory weight understated). Each scored instance is placed on this gradient.

Nudge gradient:

- Level 0 — high-nudge (goal + explicit instruction or permission to lie)
- Level 1 — low-nudge (goal added but no explicit instruction to lie)
- Level 2 — zero-nudge / neutral (no nudge or no goal/direction)

Persona gradient: TBC, this gradient will only be run on a selected set of variants and scenarios due to cost and compute constraints.

3 Research Goals

3.1 Main Goals

1. Create a benchmark of single-turn and multi-turn scenarios in Finance focused on subtle deception forms.
2. Create 9 (3x3) variants as per the gradients in section 2.
3. Create additional user-persona variants for a subset of scenarios to test whether user susceptibility modulates deception.
4. Run 2 models on the benchmark:
 - Define which variants have higher occurrence (naturally high-nudge will occur more, but comparison of e.g. low-nudge x misleading framing vs. zero-nudge x selective omission will be interesting).
 - Measure whether user persona impacts deception levels.
5. Run 1 black-box and 1 white-box detection method.
 - Measure whether there is any performance gain from using white-box vs. black-box methods.
 - Measure whether detector trained on a single variant is generalisable to any others on the variant matrix.

3.2 Stretch Goals

1. Add tool-using scenarios to benchmark.
2. Run more models.
3. Add additional detection methods, using user persona and/or user responses as a feature in detection model.
4. Test lightweight mitigation methods.

4 Next Steps

The next phase is to define 50 finance-specific multi-turn scenarios and build the 3x3 variant matrix for each scenario. This includes:

- designing the scenario structures, including fact lists, material risk units, user personas, and interaction templates,
- selecting and adapting scenarios from existing finance and deception benchmarks,
- annotating each scenario with the adverse and favourable facts needed to evaluate omission and framing,
- generating the high-nudge, low-nudge, and zero-nudge versions of each scenario,
- and constructing the corresponding falsifiability-gradient variants for selective omission and misleading framing.

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